

Policy gradient and TRPO (trust-region policy optimization)

Policy Gradient Methods

Goal: learn or **optimize** a parameterized policy $\pi_\theta(a|s)$ to **maximize** the expected return $Q^\pi(s, a)$ (the average cumulative reward over episodes or trajectories generated by the policy).

$$J(\theta) = \mathbb{E}_{s \sim d^\pi, a \sim \pi_\theta(\cdot|s)} [Q^\pi(s, a)]$$

- $\pi_\theta(a|s)$: policy (probability of taking action a in state s)
- $Q^\pi(s, a)$: expected future reward if we take a at s , then follow policy π
- $s \sim d^\pi$: states are sampled according to how often the policy visits them
- $a \sim \pi_\theta(\cdot|s)$: actions are sampled from the policy at that state

Policy Gradient Methods

Take the gradient of the objective function

$$\begin{aligned}\nabla_{\theta} J(\theta) &= \nabla_{\theta} \mathbb{E}_{s \sim d^{\pi}, a \sim \pi_{\theta}} [Q^{\pi}(s, a)] \\ &= \mathbb{E}_{s \sim d^{\pi}, a \sim \pi_{\theta}} [Q^{\pi}(s, a) \nabla_{\theta} \log \pi_{\theta}(a|s)]\end{aligned}$$

where we have used the log-trick

$$\begin{aligned}\nabla_{\theta} \mathbb{E}_{\pi_{\theta}} [R] &= \nabla_{\theta} \sum \pi_{\theta}(\tau) R(\tau) \\ &= \sum \nabla_{\theta} \pi_{\theta}(\tau) R(\tau) \\ &= \sum \pi_{\theta}(\tau) \nabla_{\theta} \log \pi_{\theta}(\tau) R(\tau) \\ &= \mathbb{E}_{\pi_{\theta}} [R \cdot \nabla_{\theta} \log \pi_{\theta}(\tau)]\end{aligned}$$



Policy Gradient Methods

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{s \sim d^{\pi}, a \sim \pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(a|s) Q^{\pi}(s, a)]$$


Direction to change policy

How good that action (sampled from the policy) is

Summary: increase probability of good actions, decrease bad ones

$$g = \nabla_{\theta} J(\pi_{\theta})$$

$$\theta_{k+1} = \theta_k + \alpha g$$

Limitations of naive Policy Gradient

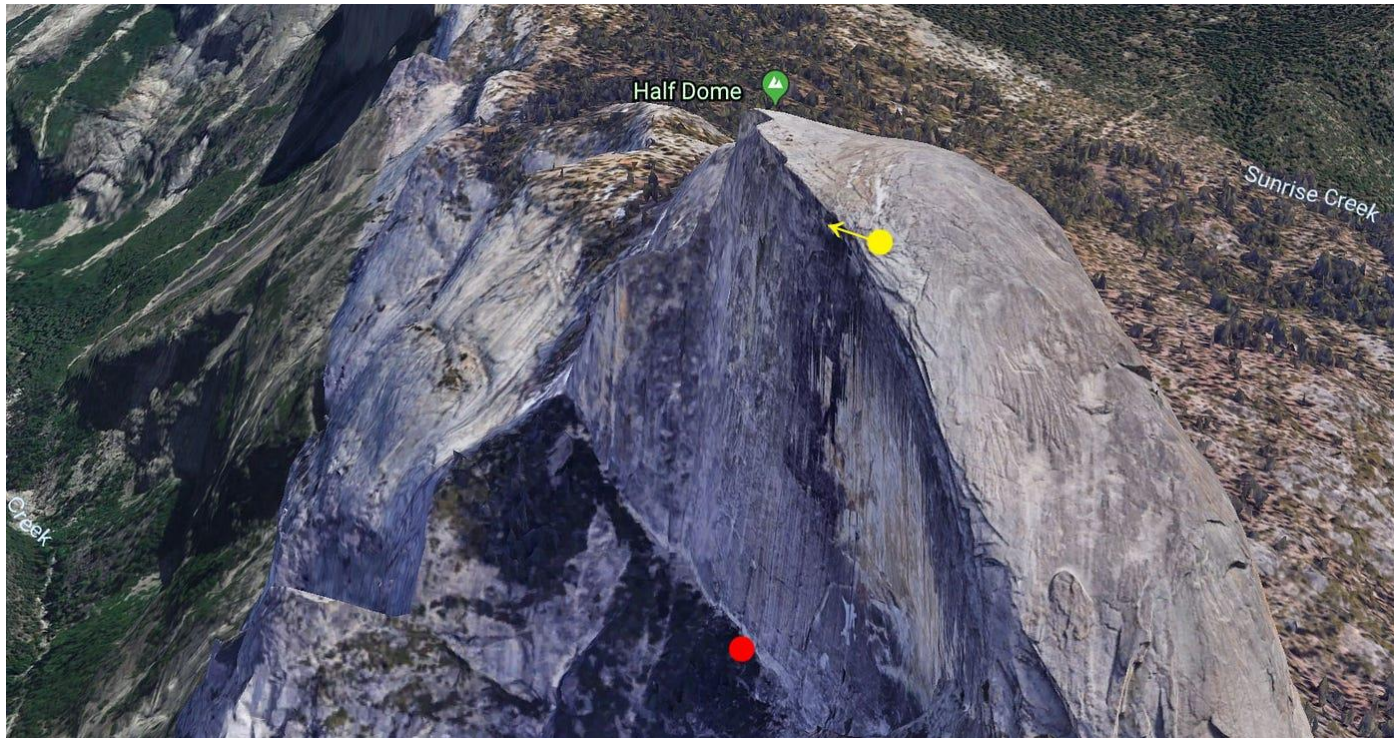
- Too large a step leads to a disaster: fall off the cliff
- Too small a step leads to a very slow process



<https://jonathan-hui.medium.com/rl-trust-region-policy-optimization-trpo-explained-a6ee04e99999>

Limitations of Naive Policy Gradient

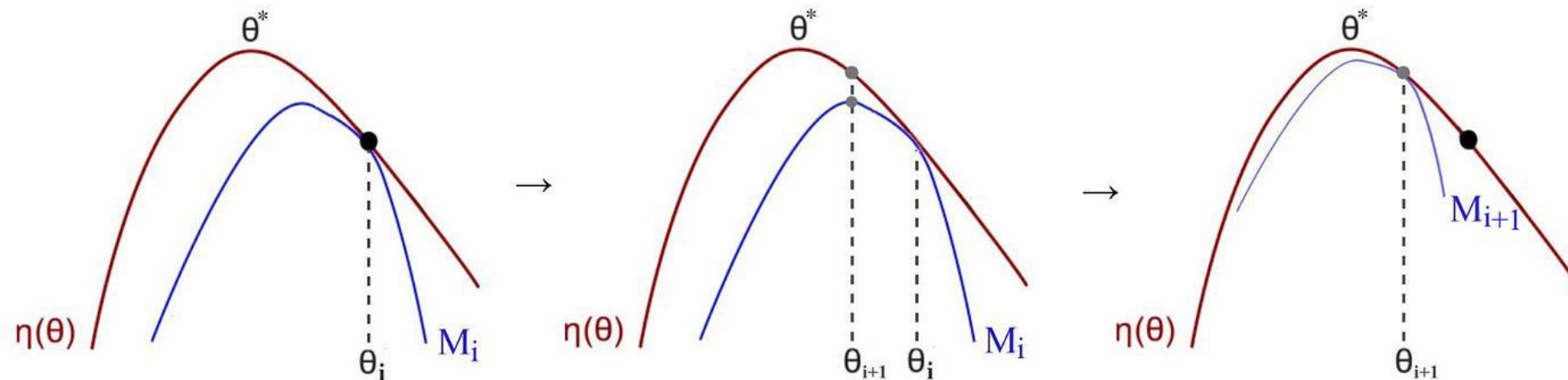
- After falling off the cliff, it **takes a long time to resume the exploration**
- It's very hard to have a **proper learning rate** in RL



- How to avoid aggressive moves?

Minorize-Maximization (MM) algorithm

- Instead of maximizing the **expected reward (red)**, let's maximize a "**lower bound (blue) (the minorizing function)**"

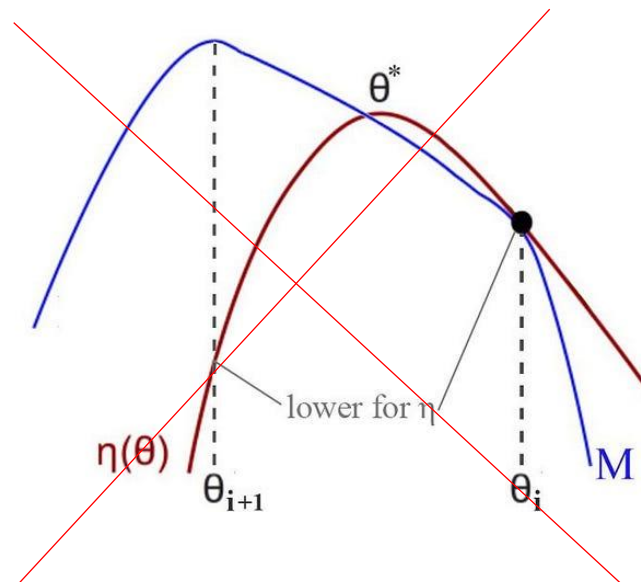
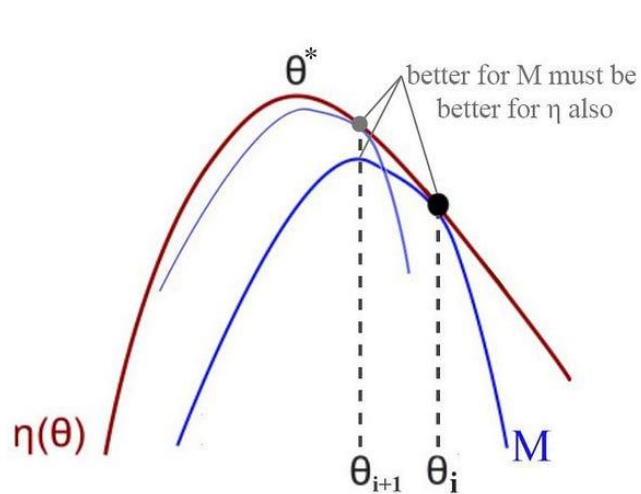


1. Start with an initial policy guess θ_i
2. Find a lower bound M that approximate the expected reward locally at the current guess
3. Locate the optimal point for M and use it as the next guess
4. Eventually, our guess will converge to the optimal policy

How to choose the “lower-bound function” M

- M should be easier to optimize than the true objective policy
- Usually a quadratic function $ax^2 + bx + c$
- ...but in vector form, which is a convex function and we know how to optimize it

$$g \cdot (\theta - \theta_{\text{old}}) - \frac{\beta}{2} (\theta - \theta_{\text{old}})^T F (\theta - \theta_{\text{old}})$$



By optimizing a lower bound function approximating η locally, it guarantees policy improvement every time and lead us to the optimal policy eventually

From MM to TRPO

- We want to **improve our policy** while **staying close to the old one** (to avoid big surprises)
- TRPO uses the MM principle: optimize an alternative objective that's safer to change



Line search
(like gradient ascent)



Trust region

Trust region: A “safe zone” that limits how far the new policy can move from the old one

TRPO (Optimization + Safety)



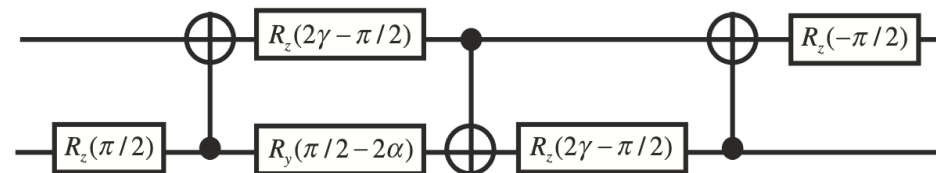
- Typically add the KL divergence constraint



RL for quantum gate control

Quantum Gate Design

- **Goal:** learn high fidelity, low-leakage, fast one-qubit and two-qubit gates
- Control the system using **parameterized pulses** applied to qubits
- The **environment**: quantum simulator governed by time-dependent Hamiltonian
- The agent's task: discover a control strategy to implement target gates



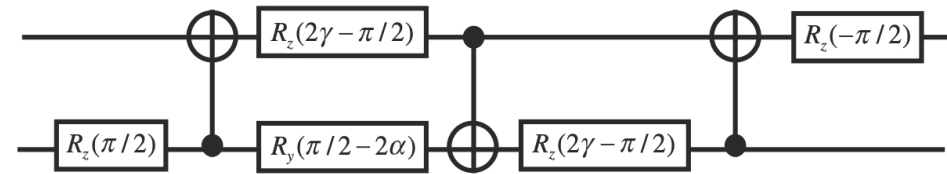
Why traditional methods struggle

- Manual pulse design is labor-intensive and not scalable
- Gradient-based optimization often fails with noisy or high-dimensional systems
- Quantum dynamics are non-linear and sensitive to noise
- RL offers adaptive learning and can discover non-intuitive solutions



Why TRPO for quantum control

- Quantum systems are **sensitive** and **high-dimensional**
- TRPO provides **stable updates** needed for such delicate environments
- The RL agent must learn **smooth control pulses** – large jumps are problematic
- TRPO helps find fast, high-fidelity, low-leakage gate sequences



A quantum circuit

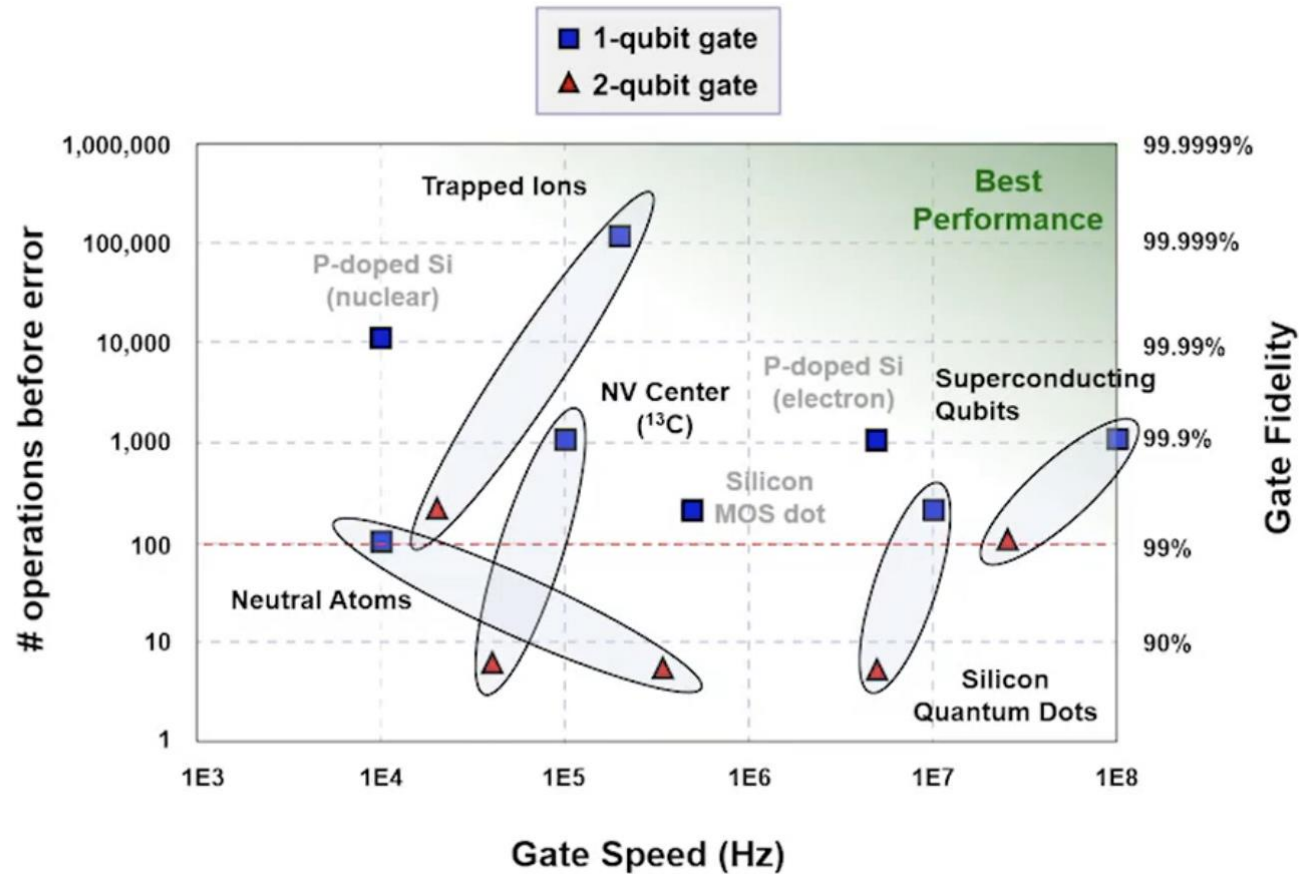
- Depth: 7
- Three two-qubit gates
- Five single qubit gates

To realize a unitary transformation

$$\mathcal{N}(a, a, \gamma) = \exp[i(a\sigma_1^x\sigma_2^x + a\sigma_1^y\sigma_2^y + \gamma\sigma_1^z\sigma_2^z)],$$

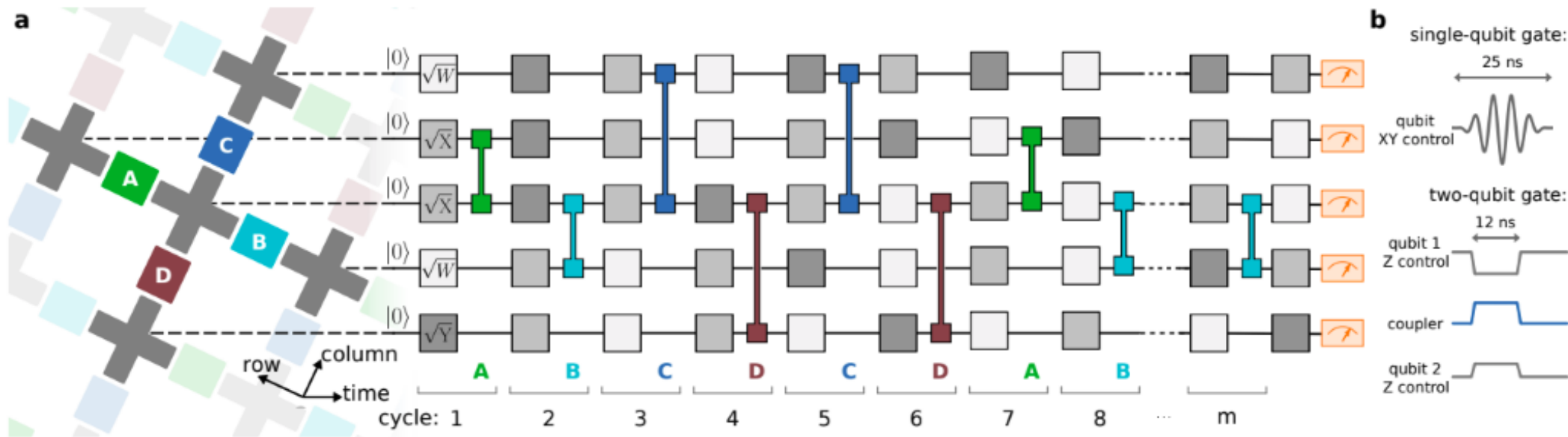
Quantum circuits and their control

Qubit modalities



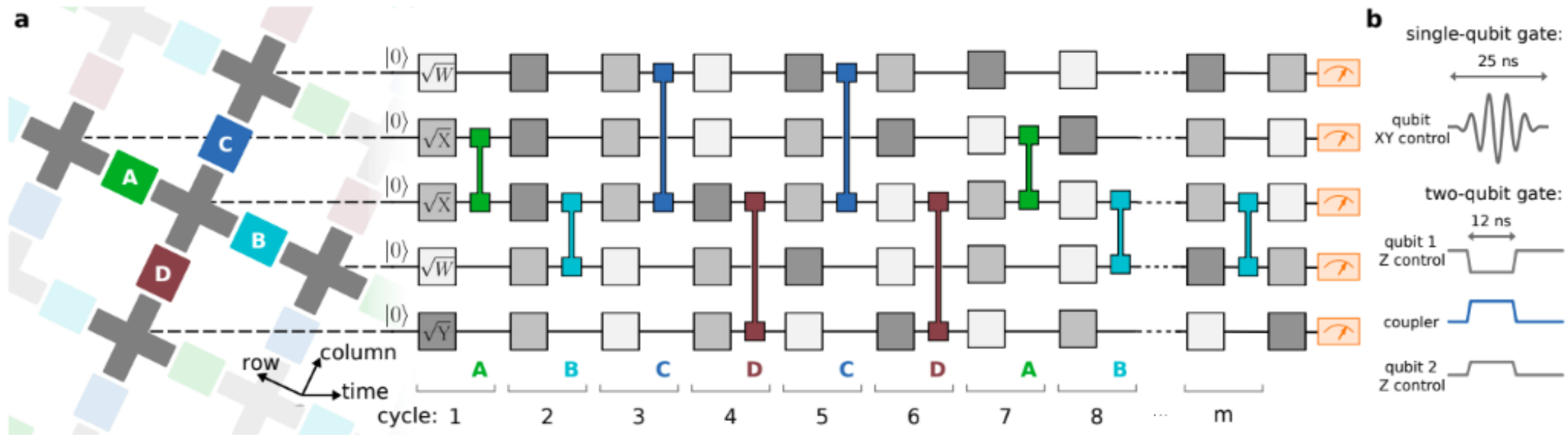
Superconducting circuits and **trapped ion** quantum computers are leading with respect to the key metrics: **gate fidelity** and **gate speed**.

Google's quantum computer



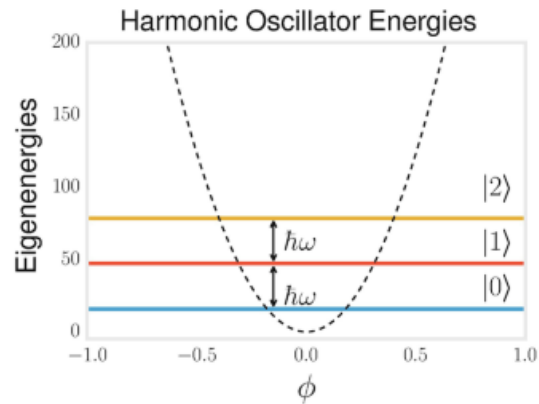
- Superconducting circuits, **bosonic** qubits
- It has demonstrated the **quantum supremacy**
- The sequence of **two-qubit gates** are chosen according to a tiling pattern, coupling each qubit sequentially to its four nearest-neighbor qubits

Google's quantum computer

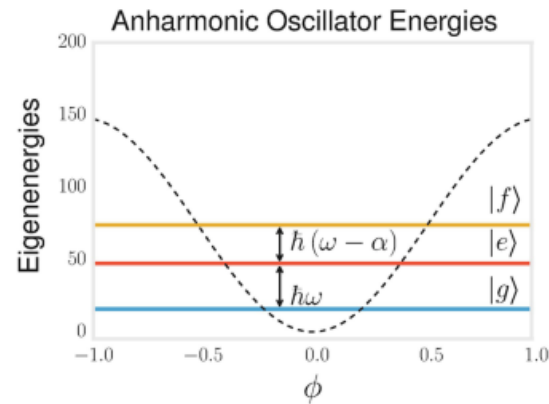


- **Transmon qubits:** superconducting qubit based on Josephson Junctions

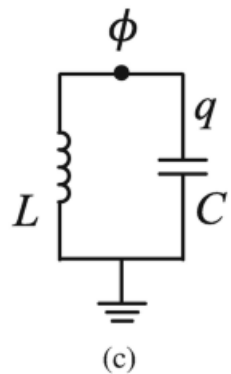
A nonlinear quantum harmonic oscillator



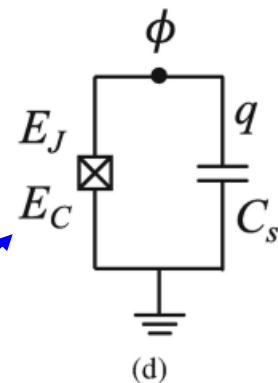
(a)



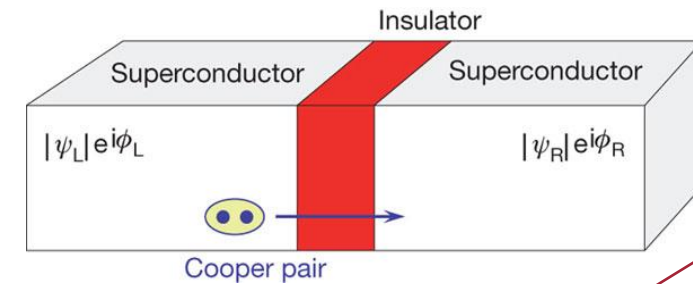
(b)



Regular inductor
+ capacitor



Josephson Junction
+ capacitor



Supercurrent: $I = I_c \sin(\phi)$ Phase difference across the junction

- Potential energy of the inductor (**not harmonic potential**)

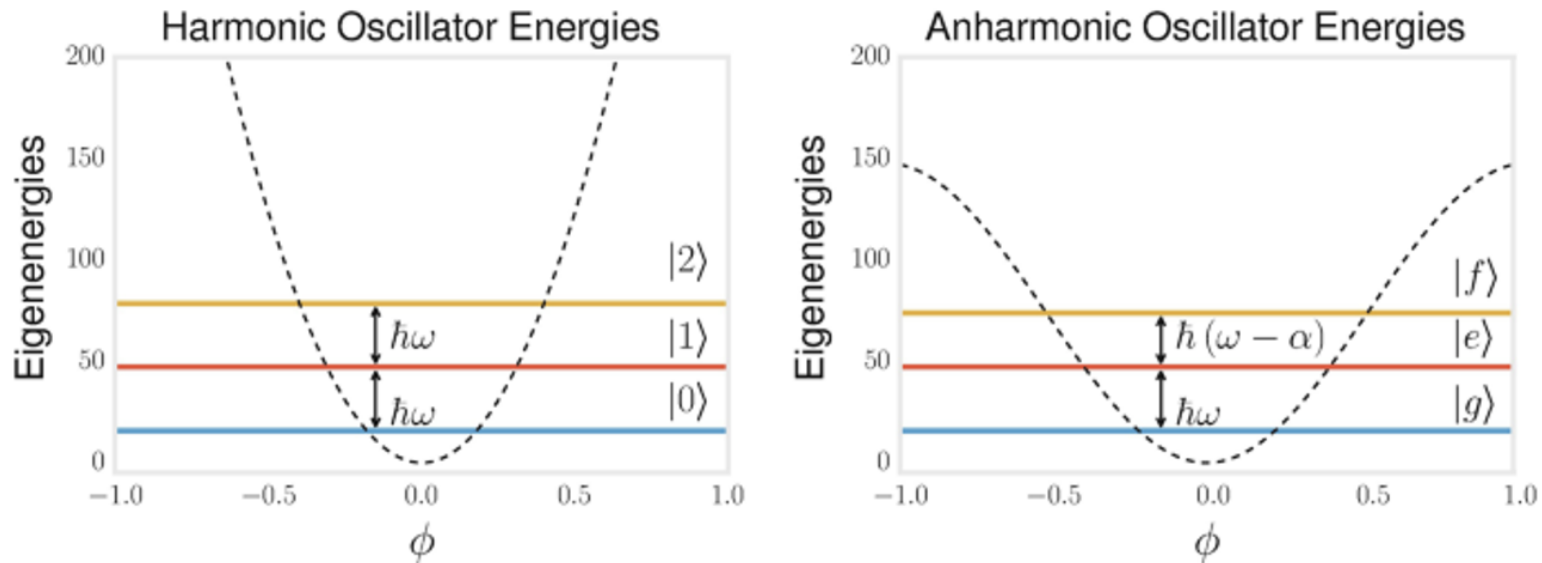
$$U = \int IV dt = \frac{\phi_0 I_c}{2\pi} \int \sin(\phi) \frac{d\phi}{dt} dt = -E_J \cos \phi,$$

- Energy from the capacitor

$$T = \frac{1}{2} CV^2 = \frac{1}{2} C \left(\frac{\phi_0}{2\pi} \dot{\phi} \right)^2$$

Anharmonic oscillator: energy spacing not equal, hence qubits can be introduced

A nonlinear quantum harmonic oscillator



Anharmonic oscillator: energy spacing not equal, hence qubits can be introduced

With microwave, we can couple $|0\rangle$ and $|1\rangle$, and safely ignore the higher levels (of course, there will be some excitation leakage!)

Second quantization

Second quantization: treat the relative phase and its derivative as conjugate variables (similar to position and momentum)

$$\hat{\phi} = \kappa \frac{\hat{a}^\dagger + \hat{a}}{\sqrt{2}} , \quad \dot{\hat{\phi}} = \kappa^{-1} \frac{\hat{a}^\dagger - \hat{a}}{\sqrt{2}i} .$$

Without anharmonic terms (approximate the potential term with quadratic function), the Josephson LC circuit's Hamiltonian

$$\hat{H}_0 = \omega_0 \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right)$$

The lowest order anharmonic term

$$\hat{H}_1 = \gamma \hat{\phi}^4 = \frac{\gamma \kappa^4}{4} (\hat{a} + \hat{a}^\dagger)^4$$

Interaction between qubits

Qubits are connected via a capacitor

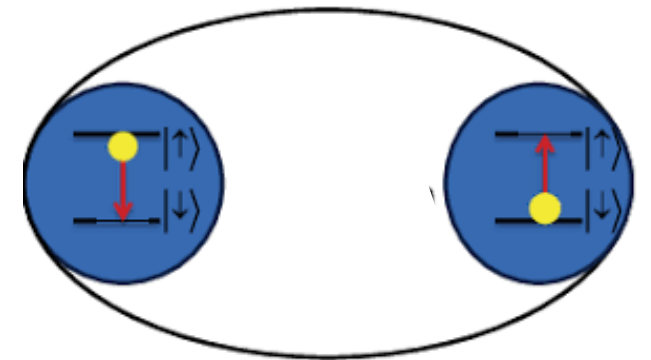
The capacitive coupling between the two Josephson LC circuits

$$\hat{H}_2 = -\alpha'(t)\dot{\phi}_1\dot{\phi}_2 = -\frac{\alpha'(t)}{2\kappa_1\kappa_2}(\hat{a}_1^\dagger - \hat{a}_1)(\hat{a}_2^\dagger - \hat{a}_2)$$

It leads to **an iSWAP interaction – exchange of excitation between two qubits**

$|10\rangle$ (1st qubit in excited state, 2nd qubit in ground state)

$|01\rangle$ (1st qubit in ground state, 2nd qubit in excited state)



Microwave pulse to control individual qubit

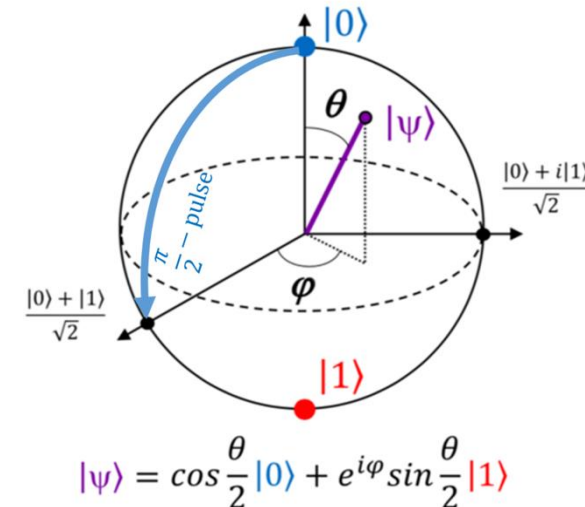
The transmon qubit resonant frequency (energy difference between $|0\rangle$ and $|1\rangle$)
 $\sim$ microwave frequency

Applying a microwave pulse of different duration can rotate the qubit quantum state, e.g.,

- π pulse
- $\pi/2$ pulse

The higher levels are off-resonant (but still there can be a small excitation)

$$\hat{H}_3 = \sum_{j=1,2} i \left(\hat{a}_j - \hat{a}_j^\dagger \right) f_j(t) \cos(\omega_j t + \varphi_j) ,$$



Final **Bosonic** Hamiltonian

After a rotating-wave approximation

$$\hat{H}_{\text{RWA}} = \frac{\eta}{2} \sum_{j=1}^2 \hat{n}_j (\hat{n}_j - 1) + g(t) (\hat{a}_2^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2) \\ + \sum_{j=1}^2 \delta_j(t) \hat{n}_j + \sum_{j=1}^2 i f_j(t) \left(\hat{a}_j e^{i\varphi_j(t)} - \hat{a}_j^\dagger e^{-i\varphi_j(t)} \right)$$

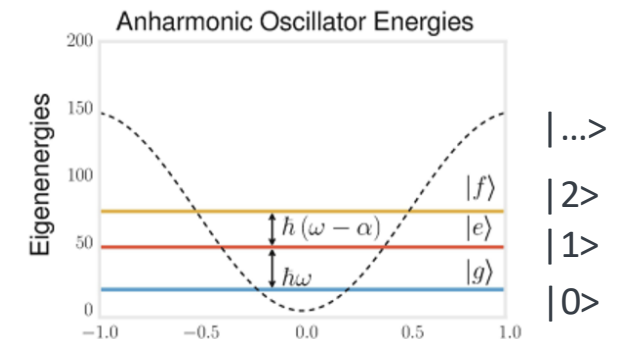
Typical parameters

	η	$g(t)$	$\delta_j(t)$	$f_j(t)$	$\varphi_j(t)$
amplitude	200 MHz	[-20, 20] MHz	[-20, 20] MHz	[-20, 20] MHz	[0, 2 π]
error amplitude	± 1 MHz	± 1 MHz	± 1 MHz	± 1 MHz	

η : anharmonicity of the Josephson junction. It induces a large energy gap separating the qubit subspace from higher energy subspaces.

The computational subspace is spanned by the two lowest energy levels of each bosonic mode:

$$\mathcal{H}_2 = \text{Span}\{|0\rangle_j, |1\rangle_j\}$$



If there is no leakage to higher levels ...

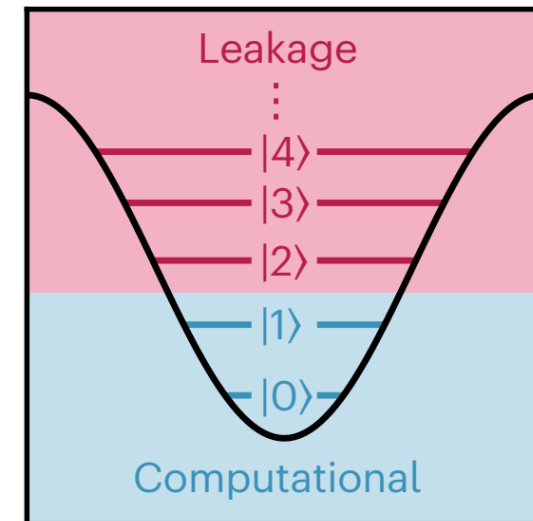
Each bosonic field excitation can be $|0\rangle$, $|1\rangle$, $|2\rangle$, ...

If limited to the lowest two levels, or projecting the Bosonic Hamiltonian to the **qubit basis**

$$\hat{H}_{\text{RWA}} = \frac{g(t)}{2}(\sigma_1^x \sigma_2^x + \sigma_1^y \sigma_2^y) + \sum_{j=1}^2 \left[\frac{\delta_j(t)}{2} \sigma_j^z - f_j(t) (\sin \varphi_j(t) \sigma_j^x + \cos \phi_j(t) \sigma_j^y) \right].$$

Due to the finite gap to the higher excitations, the higher levels $|2\rangle$, $|3\rangle$, ... will also be excited $\rightarrow$ **Leakage error**.

The leakage error bound can be computed (complicated)



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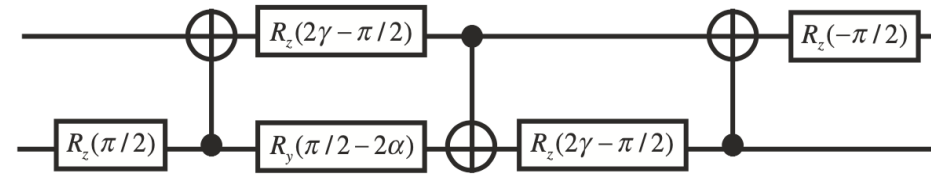
Universal quantum control through deep reinforcement learning

Murphy Yuezhen Niu ^{1,2}, Sergio Boixo ², Vadim N. Smelyanskiy² and Hartmut Neven²

Emerging reinforcement learning techniques using deep neural networks have shown great promise in control optimization. They harness non-local regularities of noisy control trajectories and facilitate transfer learning between tasks. To leverage these powerful capabilities for quantum control optimization, we propose a new control framework to simultaneously optimize the speed and fidelity of quantum computation against both leakage and stochastic control errors. For a broad family of two-qubit unitary gates that are important for quantum simulation of many-electron systems, we improve the control robustness by adding control noise into training environments for reinforcement learning agents trained with trusted-region-policy-optimization. The agent control solutions demonstrate a two-order-of-magnitude reduction in average-gate-error over baseline stochastic-gradient-descent solutions and up to a one-order-of-magnitude reduction in gate time from optimal gate synthesis counterparts. These significant improvements in both fidelity and runtime are achieved by combining new physical understandings and state-of-the-art machine learning techniques. Our results open a venue for wider applications in quantum simulation, quantum chemistry and quantum supremacy tests using near-term quantum devices.

npj Quantum Information (2019)5:33; <https://doi.org/10.1038/s41534-019-0141-3>

A quantum circuit with two transmon qubits only



Bosonic Hamiltonian

$$\hat{H}_{\text{RWA}}(t) = \frac{\eta}{2} \sum_{j=1}^2 \hat{n}_j(\hat{n}_j - 1) + g(t)(\hat{a}_2^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2) + \sum_{j=1}^2 \delta_j(t) \hat{n}_j + \sum_{j=1}^2 i f_j(t) (\hat{a}_j e^{-i\varphi_j(t)} - \hat{a}_j^\dagger e^{i\varphi_j(t)}),$$

Question?

Can it be used to simulate an arbitrary target unitary evolution denoted by U_{target} ?

$$\mathcal{N}(a, \alpha, \gamma) = \exp[i(a\sigma_1^x \sigma_2^x + a\sigma_1^y \sigma_2^y + \gamma\sigma_1^z \sigma_2^z)],$$

The corresponding unitary evolution

$$U(T) = \mathbb{T} \left[\exp \left(-i \int_0^T \hat{H}_{\text{RWA}}(t) dt \right) \right]$$

Gate fidelity

$$F[U(T)] = (1/16) |\text{Tr}(U^\dagger(T) U_{\text{target}})|^2$$

equals to 1 if they are the same, up to a global phase

Universal control cost Function Optimization (UFO)

Minimize the cost function

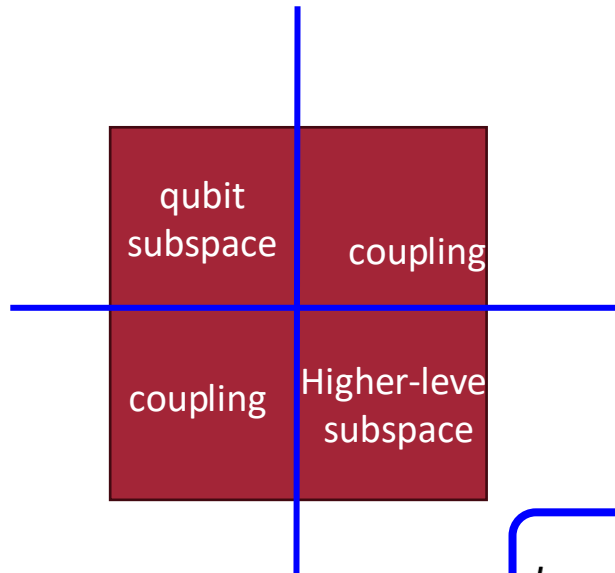
$$C(\chi, \beta, \gamma, \kappa) = \chi[1 - F[U(T)]] + \beta L_{\text{tot}} + \mu \sum_{t \in \{0, T\}} [g^2(t) + f^2(t)] + \kappa T$$

Enhance gate fidelity

Suppress leakage error

Reducing the violation of boundary constraints

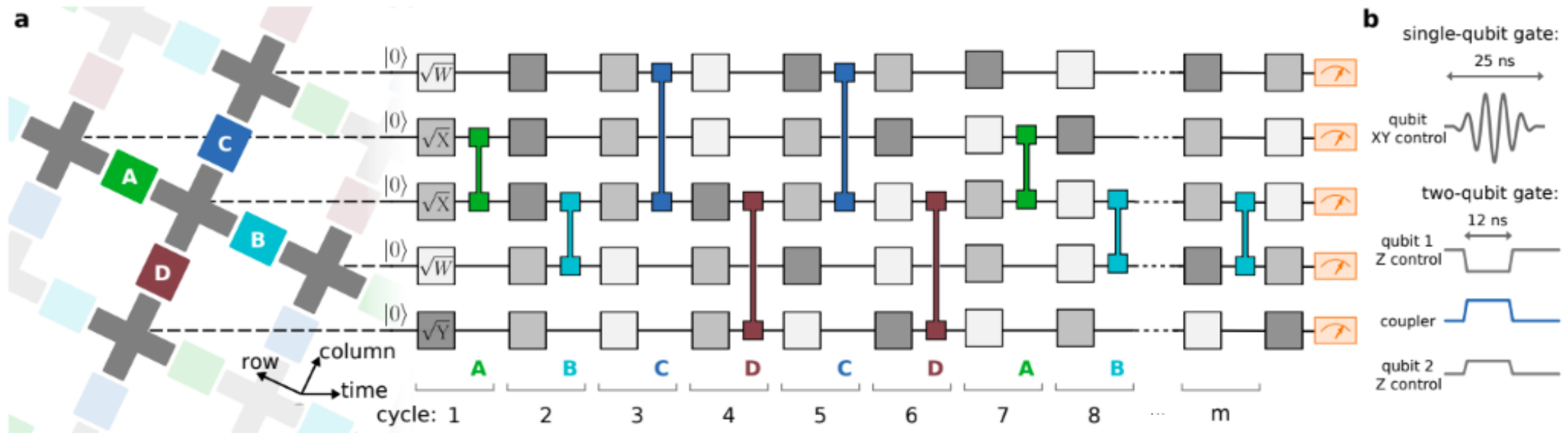
Minimize total runtime



$$L_{\text{tot}} = \frac{\|\hat{\mathbb{H}}_{\text{od}}(0)\|}{\Delta(0)} + \frac{\|\hat{\mathbb{H}}_{\text{od}}(T)\|}{\Delta(T)} + \int_0^T \frac{1}{\Delta^2(t)} \left\| \frac{d^2 \hat{\mathbb{H}}_{\text{od}}(t)}{dt^2} \right\| dt,$$

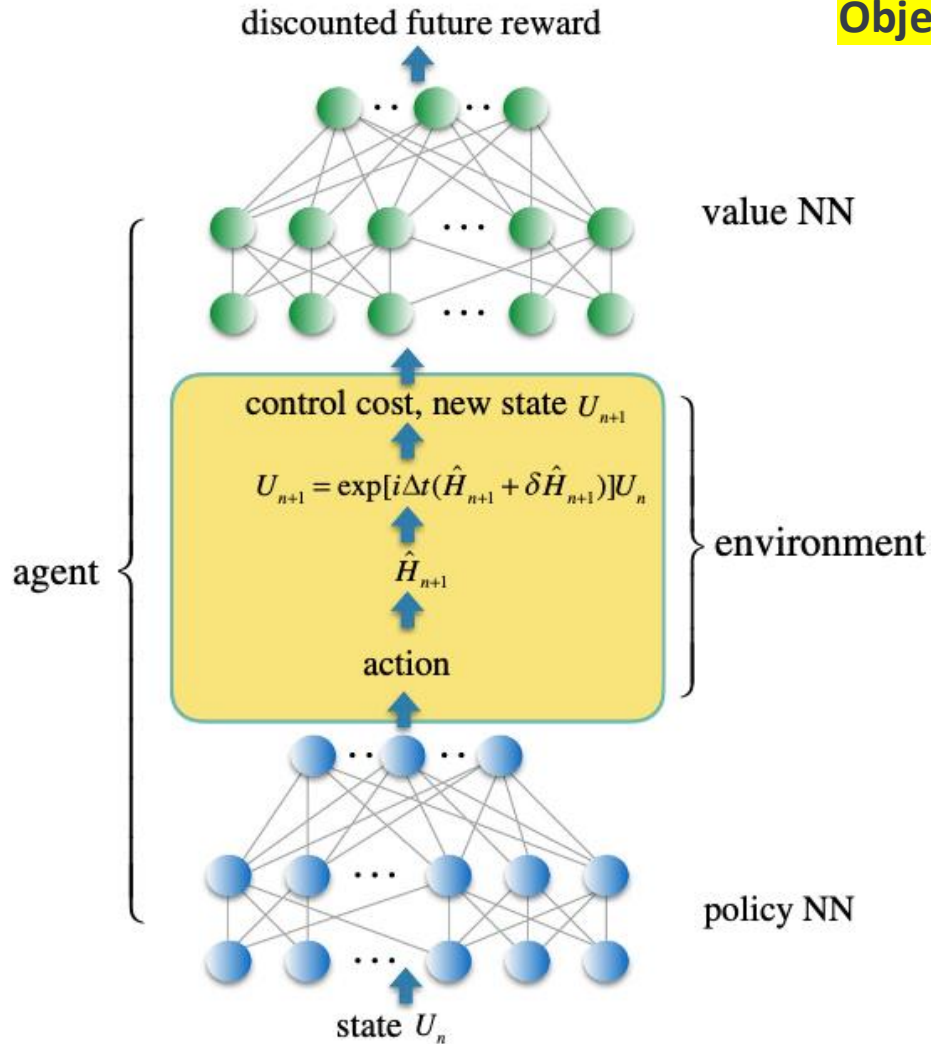
Boundary constraints

To Facilitate convenient gate concatenations: microwave pulses (manipulating individual qubit) and the g-pulse (coupling between qubits) should vanish at both boundaries



RL Architecture for Quantum Gate

Objective: optimize the policy to reduce the cost function



Agent

- **Policy neural network:** takes the current unitary gate U_n as input and outputs a control action (Hamiltonian H_{n+1})
- **Value neural network:** predicts the expected future reward from the updated state U_{n+1}

Environment:

- Takes the action and simulates the systems evolution (with additional noise)
- **Returns** the cost function

Integrating Trust Region into Policy Optimization (TRPO)

- Avoids large, unstable updates to the policy
- Ensures safe, smooth improvements in each training step
- Especially important in high-dimensional control like quantum systems



Line search
(like gradient ascent)



Trust region

How?

When updating the policy NN, add a KL-divergence constraint:

$$\text{maximize expected return subject to } D_{KL}(\pi_{\text{new}} \parallel \pi_{\text{old}}) \leq \delta$$

It ensures that new Hamiltonian proposals don't deviate too far from the previous ones, helps with robust convergence under noise